Upper respiratory tract infections

1) Which of the following bacteria is the most frequent cause of pharyngitis ?

a) Streptococcus pneumoniae b) Staphylococcus aureus
c) Streptococcus pyogenes d) Mycoplasma pneumonia

2) Which of the following bacteria causes otitis media ?

a)Streptococcus pneumoniae b) Streptococcus agalactiae
c) Streptococcus pyogenes d) Streptococcus intermedius

3) Which of the following cocci is a part of normal flora in the mouth ?

a) Streptococcus agalactiae b) Staphylococcus saprophyticus
c) Staphylococcus epidermidis d) Streptococcus mutans

4) What are the different infections caused by Group A Streptococcus spp ? Select all the correct answers

a) Pharyngitis b) Impetigo

c)Scarlet fever d) Meningitis

5) Which of the following is the complications of autoimmune reaction from pharyngitis ?

a) Rheumatic fever b) Scarlet fever

c) Fasciitis d) None of the above

6) Which of the following is NOT the virulent factor , an important antigenic property of S. pyogenes ?

a) endotoxins b) Streptolysin

c) Streptokinase d) M protein

7) Which of the following condition is non - suppurative sequelae of Streptococcus pyogenes infections ?

a) Acute rheumatic fever b) Acute glomerulonephritis

c) Erythema nodosum d) All of these

8) Which of the following cross - reacts with cell wall carbohydrate of Streptococca pyogenes ?

a)Cardiac valves b) Synovial fluid

c) Myocardium d) Vascular intima

9) Streptococcus pyogenes can be differentiated from other haemolytic Streptococci on the basis of :

a) Bacitracin sensitivity b) Erythromycin sensitivity

c) Aminoglycosides sensitivity d) Penicillin sensitivity

10) Streptolysin O is :

a) Antigenic b) oxygen - labile

c) heat - labile d) all of these

11) Which of the following test (s) may establish retrospective diagnosis of streptococcal infection ?

a) Anti - DNase B test b) ASO test

c) Streptozyme test d) All of these

12) The bacterial sore throat is caused by :

a)Staphylococcus aureus b) Haemophilus spp
c)Streptococcus pyogenes d) Mycoplasma pneumonia

13) Streptococus pyogenes comprises which Lancefield group ?

a) A b) B c) C d) D

14) Which of the following immunity is protective against infection S.pyogenes ?

a) Cell - mediated immunity b) Antigen - antibody immunity
c) Humoral immunity d) No cure present

15) In which stage of pertussis is the characteristic whooping sound made ?

a) Convalescence b) Catarrhal

c) Paroxysmal d) Prodromal

16) _____ is a common cause of sore throat and often infects the conjunctiva as well as the pharynx to cause pharyngoconjunctival fever .

a) Adenovirus b) Epstein - Barr virus

c) Cytomegalovirus d) Enterovirus

17) All of the following statements regarding Corynebacterium diphtheriac are true , except ?

a) It can be identified by using tests for toxigenicity

b) The toxin inhibits protein synthesis

c) Toxin has adverse effects on cardiovascular and nervous systems

d) Native chromosome is responsible for toxin production

18) The diphtheria toxin is a type of

a) Protein A - B toxin b) Lipid A - D toxins

c) Carbohydrate A - B toxin d) Lipid A - B toxins

e) Protein A - D toxin

19) Which of the following is NOT a characteristic of diphtheria in an infected person ?

a) Fatigue b) Fever and chills

c) Pseudomembrane in the back of the throat

d) Difficulty urinating e) Bull - neck appearance

20) What portion of the diphtheria toxin inhibits protein synthesis within an infected cell ?

a) E portion b) D portion c) C portion

d) B portion e) A portion

21) The culture medium for Coryneacterium diptheriae is?

a) Loeffler's serum slope b) MacConkey agar

c) Sabouraud's agar d) Lowenstein - Jensen medium

22) Diphtheria toxin acts by ?

a) Inhibiting acetylcholine release

b)Inhibiting glucose transport

c)Increasing levels of cyclic AMP

d) Inhibiting protein synthesis

23) One of the following statements regarding diphtheria toxin is true :

a) Toxin is produced by all corynebacteria

b) Positive Schick test suggest immunity

c) Toxin production is influenced by critical concentration of iron

d) It is an endotoxin

24) Elek's gel preciptiation test is for ?

a) Gonococcus b) Diphtheria

c) Hemophillus d) Anthrax

25) Schick's test is used to determine?

a) Suseptibility of the patient of diphtheria

b) Immunity acquired against C diphtheriae

c) Hypersensitivity to C diphtheriae

d) None of the above

26) Which of these small , gram - negative microorganisms are important human pathogens ?

a) Haemophilus influenzae b) Bordetella pertussis

c) Haemophilus parainfluenzae d) A , B and C

27) was one of the leading bacterial pathogens that caused severe upper respiratory tract infections and even death in children during the nineteenth century , it is now rare in countries where routine vaccination is given to children.

a) Haemophilus influenzae b) Bordetella pertussis

c) Haemophilus influenzae type b d) Francisella tularensis

28) Which of the following do not prove to be helpful for the treatment of whooping cough ?

a) Cough syrups and expectorants

b) DPT vaccine .

c) Macrolides

d) None of the above

29) A diagnosis of diphtheria is confirmed by :

a) Microscopic appearance of organisms stained with methylene blue

b) Isolation of a typical colony on Tinsdale's agar

c) Isolation of typical organisms from materials such as blood , showing invasiveness

d) Detection of 3 phage plaques in cultures of suspicious isolates

e) Demonstration of toxin production by a suspicious isolate .

30) Pertusis vaccine is a :

a) Live vaccine b) Killed vaccine

c) Toxoids d) None of the above

31) Which of the following is true of Haemophilus influenzae ?

a) Invasive infections are most commonly associated with encapsulated strains

b) Most invasive infections occur in infants during the neonatal period

c) Most human infections are acquired from domestic pets

d) The organism can be readily cultured on sheep blood agar in an environment of elevated CO2

32) Most common complication of pertusis is :

a) Encephalopathy b) Seizures

c) Epistaxis d) Pneumonia

33) A virus such as influenza which emerges suddenly and spreads globally is called :

a) Epidemic b) Endemic

c) Pandemic d) Zoonotic

34) The common cold is caused most often by :

a) Adenovirus b) Influenza virus

c) Respiratory syncytial virus d) Rhinovirus

e)Parainfluenza virus

35) Influenza virus infections can be characterized by which of the following statements ?

a) They have a high case fatality rate

b) They confer type - specific immunit

c) They frequently are transmitted from infected household pets

d) They commonly persist in human carriers

36) Coronaviruses are recognized by club - shaped surface projections that are 20 nm long and resemble solar coronas . These viruses are characterized by their ability to :

a) Grow well in the usual laboratory cultured cell lines

b) Cause the common cold

c) Infect infants ore frequently than adults

d) Grow profusely at 34 ° c

37) Antigenic variation in influenza virus is a phenomenon that:

a) Occurs by the addition of new antigenic determinants

b) Is more prominent in b strains than in a strains

c) Affects both the haemagglutinin and neuraminidase antigens

d) Has no impact on the clinical management of influenza

38) the serologic grouping of streptococci is a function of which of the following cellular components ?

a) C carbohydrate b) M protein c) T protein

d) Hyaluronic acid e) Teichoic acid

Answers for upper respiratory tract infections MCQ

1- c 2 – a 3 - d 4 - a , b , c , 5 - a 6 - a

7 - d 8 - a 9 - a 10 - d 11 - d 12 - c

13 - a 14 - c 15 - c 16 - a 17 - d 18 - a

19 - d 20 - e 21 - a 22 - d 23 - c 24 -b

25 - a 26 - d 27 - c 28 - a 29 – e 30 - d

31 - a 32 - d 33 - c 34 - d 35 - b 36 - b

37 - a , c 38 – a

Lower respiratory tract infections

1) All of the following are the characteristics of Streptococcus pneumoniae , EXCEPT?

a) The colonies are diplococcus and encapsulated

b) Alpha hemolytic colonies appear in blood culture media

c) It is present as normal flors in healthy individuals

d) It grows well in a high concentration of glucose media

2) Which of the following is a major virulence factor of Streptococcus pneumoniae ?

a) polysaccharide capsule b) M protein

c) streptolysin O d) streptolysin S

3) A 22 - year - old woman complains of a persistent nonproductive cough fever of 101 ° F that came on slowly over the last 4 days . Physical examination reveals some rales in the left lung base . A patchy infiltrate is seen on chest ray . She works as a secretary in a law office and has not traveled recently She is not immunocompromised and has not been hospitalized recent sample of her serum agglutinates red blood cells at 4 ° C but not at Which one of the following BEST describes the organism that is the M likely cause of her disease ?

a) A filamentous gram - positive rod that is weakly acid - fast

b) A very small bacterium that has no cell wall

c) A gram - negative diplococcus with a large capsule

d) A spirochete that has never been grown on blood agar

4)Tubercular Granulomas are made up of?

a) damaged tissue cells and bacilli only

b) fibrous and damaged tissue and bacilli only

c) necrotic tissue and damaged tissue and bacilli only

d) organized aggregates of immune cells that surround foci of infected tissues

5) Which of these microbes causes " walking pneumonia " (atypical pneumonia)

a) Klebsiella pneumoniae b) Streptococcus pneumoniae

c) Mycoplasma pneumonia d) Chlamydophila pneumonis

6) Which of the following is the most common cause of bronchiolitis ?

a) Respiratory syncytial virus b) Parainfluenza virus

c) Mycoplasma d) Adenovirus e) Streptococcus

7) Community acquired pneumonia is most commonly associated with?

a) Staphylococcus b) Listeria c) Legionella

d) Pneumococcus e) Pseudomonas

8) Acute tracheobronchitis is most commonly associated with which o following pathogens ?

a) Pseudomonas b) Klebsiella c) Influenza A Virus

9) What is a Quellung reaction?

a)The capsule detection test

b)The serological antibody detection test

c) Direct antigen detection test d) The bacterial motility test

10) Primary tuberculosis in which inhaled droplet nuclei are deposited in the peripheral respiratory alveoli , most frequently occurs in lobe(s) of the lungs .

a) Upper b) Upper and middle

c) Middle and lower d) Lower

11) MTB can produce large quantities of , which is uncommon to other mycobacteria .

a) Tuberculin b) Niacin c) Endotoxin d) Exotoxin

12) Influenza has the greatest virulence and epidemic spread

a) A b) B c) C

13) …….. has 15 recognized subtypes and is important for attachment

a) Neuraminidase b) Hemagglutinin

c) Glycoprotein d) Cilin

14) …….. is a " slower process " that may result in as little as a single mutation in the viral RNA , occuring every few years in influenza A

a) Antigenic shift b) Antigenic drift c) Bottleneck effect

15) The most common mode of transmission for influenza virus?

a) Fecal - oral route b) Inhalation of airborne droplets

c) Direct droplet spread d) Direct physical contact

16) False regarding Pneumococcus is ?

a) Causes mild form of meningitis

b) Respiratory tract of carriers is the most important source of infection

c) Common t cause of otitis media

d) Capsule is important virulence

17) Typical bacterial pneumonia is caused by :

a) Mycoplasmal pneumonia .

b) Legionnaires pneumonia .

c) Pneumocystis carinii .

d) Staphylococcus aureus .

18) All of the following can be used to diagnose atypical CAP EXCEPT ?

a) ESR and Creactive protein . b) Xrays

c) Gram stain . d) All the above

19) Mycobacteria are acid - fast positive bacteris because of :

a) the presence of lipopolysaccharide in the bacterial cell wall

b) the presence of mycolic acid in the bacterial cell wall

c) the presence of lipids

d) Both B and C options given above

20) Which of the following first - line antibiotics are usually resistant to Mycobacterium tuberculosis ?

a) Isoniazid and ciprofloxacin

b) Isoniazid and Rifampin

c) Rifampin and ciprofloxacin

d) Rifampin and streptomycin

21) What is the mechanism responsible for antibiotic resistance in Mycobacteria tuberculosis ?

a) Mutations in DNA gyrase gene

b) Alterations in beta - lactamase

c) Mutations in the catalase - peroxidase gene

d) Alterations in RNA polymerase

22) All of the following are the examples of selective media used for culture of Mycobacterium tuberculosis , except ?

a) Inspissated egg media

b) Middlebrook 7H10 / 7H11 media with antibiotics

c) Dubos media

d) Middlebrook 7H10 media without antibiotic

23) Which of the following statement true about the tuberculin test and purified protein derivative (PPD) ?

a) The presence of intradermal skin induration is observed in 6 to 8 hours after being applied

b) The redness of skin or erythema is also measured while reading the tuberculin test

c) A positive tuberculin test means that a person was infected with M. tuberculosis in the I and continues to carry the viable organism

d) A positive PPD test indicates that a person can never be infected with M. tuberculosis

24) Which of the following is fast growing , non - chromogenic rod bacilli ?

a) Mycobacterium ulcerans b)Mycobacterium avium complex
c) Mycobacterium leprae d) Mycobacterium fortuitum

25) All of the following are the symptoms of pulmonary tuberculosis , Except?

a) Weakness and fatigue .

b) Decreased body temperature

c) Weight loss .

d) Severe prolonged cough with sputum or blood

26) Which of these following categories of people do not require the test for TB infection ?

a) A 30 years old man traveling from india to the USA for the first time

b) A 45 years old woman contradicted with HIV infection

c) A 7 years old healthy girl living together with her grandmother who has tuberculosis infection

d) None of the above

27) Which of the following bacteria causes lung infection and is the most com non - tuberculous mycobacterial infection associated with AIDS patients ?

a) Mycobacterium avium complex b) Mycobacterium leprae
c) Mycobacterium gordonae d) Mycobacterium gastri

28) A 40 years old patient who has clinical symptoms pulmonary tuberculos the past two months . The chest x - ray examination revealed a typical feat tuberculosis infection . A photochromogenic (orange pigment when exposed t light) acid - fast rod bacterium was isolated from the sputum sample . The iden bacteria is :

a) Mycobacterium avium b) Mycobacterium tuberculosis
c) Mycobacterium kansasii d) Mycobacterium leprae

29) Other than Mycobacterium tuberculosis which of the given bacteria e tuberculosis infection in animals and can be transmitted to huma consumption of milk and other animal products ?

a) Mycobacterium ulcerans b) Mycobacterium leprae
Mycobacterium bovis d) Mycobacterium abscessus

30) The treatment regimen for initial therapy of tuberculosis caused by Mycobacterium tuberculosis includes ?

a) Streptomycin and rifampin

b) Isoniazid , streptomycin , and ethambutol

c) Rifampin , isoniazid , and ciprofloxacin

d) Isoniazid , rifampin , pyrazinamide , and ethambutol

31) What is the interferon - gamma release assay ?

a) The antigen detection test for the Mycobacterium spp

b) The DNA detection test in Mycobacterium spp

c) The test used as an alternative tuberculin skin test in latent tuberculosis

d) The test used as an alternative tuberculin skin test in active tuberculosis.

32) Cold agglutinin test is useful for the diagnosis of :

a) Mycoplasma pneumonia b) H. influenza

c) N. menigitidis d) Cryptococcus neoformans

33) Pneumococci can be differentiated from viridans streptococci by a following except :

a) Alpha hemolysis on blood agar b) Optochin sensitivity

c) Bile solubility d) Quellung reaction .

Answers for lower respiratory tract infections MCQ

1 - d 2 – a 3- b 4 - d 5 - c 6 - a

7 - d 8 - c 9 - a 10 - a 11 - b 12 - a

13 – b 14 - b 15 – b 16 – a 17 - d

18- c 19 - d 23 – c 24 - d 25 - b

26 - d 27- a 28 - c 29 - c 20 - b

21 - c 22 - d 30 - d 31 - a 32-a

33 -a